

Research Paper

The influence of Mars' continental-scale topographic structures on its hemispheric asymmetry in baroclinic wave activity

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ABSTRACT

Martian extratropical baroclinic waves exhibit a pronounced hemispheric asymmetry, with significantly more vigorous activity in the Northern Hemisphere (NH) compared to the Southern Hemisphere (SH). This study employs the NASA Ames Mars Global Climate Model (MGCM) to investigate the role of topography in driving this asymmetry. Through a series of simulations with varying topographic configurations, we demonstrate that continental-scale impact basins, lowlands, and highlands have a substantial influence on the intensity and distribution of baroclinic waves. Our findings reveal that the SH's relatively weaker wave activity is partly attributable to the presence of the Hellas and Argyre impact basins, which act as barriers to baroclinic wave propagation. The hemispheric asymmetry is additionally amplified by an increased wave activity in the NH due to an intensification of baroclinic waves at the boundaries of Tharsis-Acidalia, Elysium-Arcadia, and Nilosyrtris-Utopia. It is determined that such highland-lowland west-east structures contribute to a weakening in wave activity above the highlands and a strengthening above the lowlands, with the influence of the latter propagating downstream to the east with a resulting net gain in wave energy. These results provide insight into how topography influences extratropical weather systems on Mars and highlight how impact basins and highlands may shape the climate of a rocky planet.

1. Introduction

Extratropical weather systems on Mars form at mid- and high- latitudes during the pre-solstice season in the winter hemisphere and last throughout winter (Ryan et al., 1978; Banfield et al., 2004), dominated by long waves (Barnes, 1980) of baroclinic origin (Barnes et al. (1993) and James et al. (1999), with peak wave activity closely following the seasonally evolving polar ice cap edge (Wang et al., 2005). First predicted by Mars global climate modeling (MGCM) studies (Barnes et al., 1993), and then confirmed mainly in Mars Global Surveyor (MGS) Thermal Emission Spectrometer (TES) data (Banfield et al., 2004), these baroclinic waves are much stronger in Mars' Northern Hemisphere (NH) than in its Southern Hemisphere (SH) (Mooring and Wilson, 2015; Battalio et al., 2018). In particular, the wave-3 (i.e., zonal (east-west) Fourier harmonic $s = 3$) baroclinic component has been linked to frontal dust storms (Wang et al., 2003), a typical feature of mid-latitude weather in the NH that has been observed regularly in MGS Mars Observer Camera (MOC) (Cantor et al., 2001) and Visual Monitoring Camera (VMC, onboard Mars Express) (Sánchez-Lavega et al., 2025) imagery. While local dust storms can be seen in the SH

storm track (Battalio and Wang, 2021), the frontal structures are rarely present in imagery of the SH.

The seasonal behavior of the low-level transient wave activity is characterized by pre-solstice and post-solstice maxima surrounding a solstitial pause when wave activity decreases. It has been documented in the NH (Barnes, 1980; Wang et al., 2005) since the Viking landers (Ryan et al., 1978). Characterization of the interannual variability of SH transient waves from observations was made possible only later when TES and later Mars Climate Sounder (MCS) temperature observations became available. Utilizing reanalysis data (that assimilates retrieved temperatures into MGCMs) based on TES (Lewis et al., 2016) and MCS retrievals (Hinson and Wilson, 2021; Hinson, 2025, NH, SH, respectively), a consistently weaker SH temperature eddy field than what is observed in the NH (by a factor of two at 2.5 km height) has been revealed.

The intensity of the extratropical storm tracks has been argued to be a key component driving dust storm frequency, as periods of intense NH wave activity overlap with periods of increased dust storm activity (Cantor, 2007; Guzewich et al., 2015), in particular the pre-solstice season with the A-type ($L_s \sim 205^\circ\text{--}240^\circ$) and the post-solstice

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season with the C-type ($L_s \sim 305^\circ\text{--}320^\circ$) dust storm period (Kass et al., 2016), where L_s is the areocentric solar longitude. The A- and C-periods exhibit regional scale (Martin and Zurek, 1993; Cantor et al., 2001) dust storms almost every Martian Year (MY) (e.g. Kass et al., 2016; Battalio and Wang, 2021; Martín-Rubio et al., 2024) that elevate atmospheric dust content at all longitudes and at a wide range of latitudes that affect the global circulation. The main expansion period of these regional dust storms that intensify in the SH is often, but not exclusively, preceded by a flushing storm (Wang and Richardson, 2015), when frontal storms rapidly (in a few sols) transport dust from the NH extratropics to the SH through one of the vast lowlands, Acidalia-Chryse or Utopia-Isidis, in strong north-south western boundary currents (WBCs) (Joshi et al., 1995) at the eastern side of orographic elevations (i.e., Tharsis, Syrtis). The cross-hemispheric dust transport on occasion may generate additional dust lifting in the SH, eventually triggering an expanding regional storm. Previous studies (Wang et al., 2005; Hinson and Wang, 2010; Wang, 2018; Hinson and Wilson, 2021) have found that regional storms are frequently preceded by an amplification of the northern hemispheric mid-latitude wave-3 transient wave, with some regional storms that are linked to cross-hemispheric dust transport through the Utopia-Isidis lowlands are being preceded by an intense wave-2 mid-latitude wave-amplification instead.

In contrast, the weak SH storm track, which has its local peaks at around $L_s = 45^\circ$ in the SH pre- and (a more substantial peak) at $L_s = 180^\circ$ in the SH post-solstice season, rarely triggers regional dust storms. Occasionally, a regional dust storm develops around $L_s = 150^\circ$, initiated in the SH at the Aonia-Solis-Valles Marineris (ASV) track (Battalio and Wang, 2019). The strongest observed ASV storms have occurred in MY 29 (Battalio and Wang, 2021) and MY 36 (Martín-Rubio et al., 2025). However, typical ASV storms barely exceed the criteria of regional-scale storms. Their spatial extent is much more localized than the enormous A- and C-period storms.

Consequently, the interannual variability and hemispheric asymmetry of transient wave activity play a crucial role in Mars' climate, being one of the fundamental drivers of regional-scale dust storms. Extreme events (e.g., high amplitude wave-3 baroclinic events) of the extratropical storm track tend to be concentrated at specific locations in either hemisphere, the vast NH lowlands being the regions of A- and C-period flushing storms, and the strongest (southern winter) SH dust storms being located in ASV, suggesting that high amplitude wave events are influenced by topography.

Topographic zonal asymmetries also fundamentally alter the circulation of the lower atmosphere on Mars, with orographic slopes weakening baroclinic growth rates (Blumsack and Gierasch, 1972), transferring baroclinic eddy kinetic energy into stationary wave kinetic energy (Yu and Hartmann, 1995), and forming WBCs. While it has been theorized that the hemispheric asymmetry in wave activity could be (at least in part) a result of the high eccentricity of Mars' orbit, most studies have argued that it is of topographic origin, resulting either from the hemispheric crustal dichotomy of Mars or from zonal asymmetries (Mulholland et al., 2016). The role of topography in the hemispheric asymmetry of baroclinic wave activity has been considered in several idealized and more complex modeling studies (Barnes et al., 1993; Hollingsworth et al., 1996; Keppen and Ingersoll, 1995; Matheson, 2000; Mulholland et al., 2016). However, the idealized topographic setups have been quite limited, considering a flat (e.g. Hollingsworth et al., 1996) or a zonally averaged topography (e.g. Mulholland et al., 2016) to elucidate the effects of topography on transient waves.

While a flat or zonally averaged topography experiment can provide qualitative insight into whether certain climate features are affected by topography, these configurations are too simple to provide insight into which specific topography components are responsible. More targeted and carefully designed experiments are necessary. Some targeted experiments have been performed in a few studies, including filling up Hellas (Ogohara and Satomura, 2008) to analyze barotropic flow

over the impact basin or a single (idealized) Tharsis-like topography to characterize western boundary currents (Joshi et al., 1995). However, such studies have not addressed the implications for baroclinic wave activity. Moreover, crude alterations to orography may lead to abrupt changes, resulting in a near-surface atmosphere that is structurally different from realistic cases. In particular, Matheson (2000) pointed out that the removal of all zonal asymmetries can lead to a trough in the latitudinal band occupied by Hellas (in their experiment 'Trough'), culminating in the formation of a very strong, wave-3 traveling wave during SH winter. However, their 'zonally symmetric' experiment did not result in a strong wave occupying the channel that formed due to zonally averaging Hellas. Consequently, a zonally averaged topography may or may not, depending on details of the model and parametrization, lead to more vigorous transient wave activity (compared to wave activity with realistic topography) and not purely be a result of zonal variations in realistic topography suppressing wave generation, but as a consequence of a strong channel flow generated by a zonally averaged Hellas. Therefore, while it has been argued that removal of all zonal asymmetries in Mars' SH topography can lead to higher transient wave activity by Mulholland et al. (2016), it is not clear whether their simulations induced a strong channel flow as in Matheson (2000), and if it did, what effect did it have on the intense wave activity they identified in the zonally averaged topography experiment.

In this paper, we analyze the impact of different topographic structures on Mars' baroclinic wave activity by performing simulations with the NASA Ames Mars Global Climate Model (MGCM). A large number of experiments are conducted with different topographic configurations, detailed in Section 2, to elucidate the influence of the continental-scale impact basins in the SH (Hellas, Argyre), the vast lowlands (Acidalia, Utopia), and highlands (Tharsis, Elysium, Syrtis Major) on the hemispheric asymmetry in baroclinic wave activity. We examine the effects of topography on baroclinic wave activity during the pre- and post-solstice seasons, as well as the solstice season. It will be shown that if Hellas, Argyre, Syrtis Major, Arabia Terra, Utopia, and Elysium are all removed, hemispheric asymmetry in wave activity may disappear, and the two hemispheres come to a near-balance in the pre- and post-solstice seasons. The paper is organized as follows. In Section 2, we describe the MGCM and the experiments. In Section 3, we present the results of the model experiments. Outcomes are discussed in Section 4, and Section 5 provides conclusions.

2. Methods

2.1. The NASA Ames Mars GCM

The NASA Ames MGCM version 3.1 uses a cubed-sphere grid and finite-volume dynamical core developed by NOAA/GFDL (Harris et al., 2021), coupled with physics packages from the NASA Ames Legacy Mars GCM (Haberle et al., 2019). The model is run without radiatively active water ice clouds. The model utilizes surface albedo maps (Putzig and Mellon, 2007) from the MGS Mars Orbiter Laser Altimeter (MOLA), and thermal inertia (Tyler and Barnes, 2014) from Viking and Thermal Emission Spectrometer observations. In our simulations, we ran the MGCM at 'c24' horizontal resolution, where each cube-face is divided into a 24×24 subgrid, with a hybrid sigma-pressure vertical coordinate system (Simmons and Burridge, 1981) spanning 56 vertical layers, having a vertical resolution of ~ 5 m near the surface and ~ 5 km near the model top (80–90 km). The model output is interpolated horizontally to a $\sim 3.75^\circ \times 3.75^\circ$ resolution longitude-latitude grid in the post-processing. The model was run for 6 MYs from a cold start with no initialized surface frost, and a prescribed atmospheric CO_2 mass corresponding to realistic Mars conditions. The CO_2 surface cycle solves a soil conduction equation with a full surface heat balance for the top boundary condition, and zero flux for the bottom boundary condition. The first 1 MY of each simulation is discarded (as model spin-ups), and only the last 5 MYs of the simulations are considered.

Table 1
Summary of topographic configurations.

Group	Name	Topographic configuration
1	Control	MOLA topography ($3.75^\circ \times 3.75^\circ$)
	zonalmean	Zonally averaged MOLA topography
	zonalNoHellas	Topography is zonally averaged after Hellas and Argyre are removed by landfill
2	NoHellas	Both SH impact basins, Hellas and Argyre, are removed by landfill
	NoTharsis	Tharsis is replaced by a meridional dual-slope structure
	NoUtoEly	Utopia is filled up, and Elysium Mons is removed
	NoSyrtis	Arabia Terra area is removed, Syrtis Major is replaced by a southward slope
	NoHUES	Hellas, Argyre, and Utopia are filled up, and Syrtis Major, Arabia Terra, and Elysium Mons are removed

While the GCM is capable of running with a few different dust schemes (see e.g., [Bertrand et al. \(2020\)](#)), we employ a prescribed, zonally averaged background dust scenario to simulate typical ‘climatological’ conditions. The latitudinal prescription of the background dust scenario is constructed from the dust scenarios of [Montabone et al. \(2015\)](#) by identifying the latitudinally and seasonally varying minimum value of the zonally averaged dust in the period MY 29–34 to filter out the influence of global and regional dust storms. The vertical prescription is given by a modified Conrath profile that varies with season and latitude ([Montmessin et al., 2004](#)). Hence, each MY of a given experiment is forced by the same, annually-repeating dust scenario. Such dust forcing is expected to suppress interannual variability and allow the characterization of the ‘climatological’ evolution of baroclinic wave activity.

2.2. Model experiments

2.2.1. Topographic configurations

We performed MGCM simulations with various topographic configurations. Our baseline **control** simulation uses a topographical map generated from MOLA data ([Smith et al., 1999](#)). Topographic alterations can be divided into two main groups in the rest of the experiments. The first group of experiments made global-scale simplifications to topography. In the second group of experiments, topographic changes were targeted at specific regions to elucidate the effects of these regions on baroclinic wave activity. The experiment IDs and the corresponding changes to topography are summarized in [Table 1](#). The reader is referred to [Appendix](#) for the details on how topographic alterations were made precisely.

We illustrate the key topographic regions (e.g., Hellas, Tharsis, Syrtis Major) on [Fig. 1a](#), and show the various topographic configurations corresponding to experiments in group 2 on panels (b–f). The aim of performing experiments with various topographic configurations is to assess the orographic effects of the various topographic features; hence, surface albedos and thermal inertia maps were not modified. We also do not account for the impact of topographic alterations on the principal axis of inertia that would result in a shift of the poles ([Bouley et al., 2016](#)) and hence of the polar caps. The initialized atmospheric CO_2 mass is the same and does not vary across experiments. Note that although the topography in experiment **NoTharsis** in some ways may reflect that of early Mars prior to the formation of the Tharsis Montes, the experiments performed in this study do not readily translate to early Mars due to the differences in atmospheric conditions (e.g., higher surface pressure, lower solar luminosity).

2.2.2. Subtleties with altering Mars’ topography

The design and testing of our topographic experiments yielded two cautionary tales. First, topographic alterations must avoid creating sharp topographic discontinuities (or cliffs) at the boundary of the modified region. Otherwise, the resulting circulation could lead to an artificial intensification of winds at such boundaries, fundamentally altering the lower atmosphere climate. Hence, in such cases, it would be hard to argue that the resulting atmospheric changes in the experiment are due to removing the targeted structure instead of the artificial

cliffs. Therefore, our constructions specify topographic alterations that remove the targeted region without creating sharp discontinuities in topography at the boundary of the modified region. This can be done in the case of impact basins; however, the removal of Tharsis and Syrtis Major required more careful attention. We replaced them with dual-slope structures to avoid topographic discontinuities in the east–west direction and at the southern edge of the modified regions. These lessons lead us not to include a ‘**NoAcidalia**’ experiment that would aim to analyze the effects of a topography with Southern Acidalia and Chryse removed (our exact configuration is provided in [Appendix](#)). Removing Acidalia/Chryse alone would have required elevating the boundaries to the level of Arabia Terra and Tharsis, while keeping the northern boundaries low to avoid discontinuities at the north. Hence, the removal of Acidalia/Chryse should be accompanied by the removal of Arabia Terra and Tharsis. Such an experiment would not explicitly provide information on the effects of Acidalia/Chryse on baroclinic waves, but rather on the combined impact of Tharsis, Acidalia, and Arabia Terra.

3. Results

We assess the influence of topography on baroclinic wave activity by considering the mean extratropical eddy kinetic energy for each season in each experiment. We first identify baroclinic waves (zonal u^* and meridional v^*) by applying a 1-sol and a 20-sol running mean difference on the zonal and meridional winds to filter out tides and quasi-stationary waves. Second, at any time and height, at each latitude-longitude gridpoint (θ, ϕ) , the area-weighted EKE (per unit mass) is given by

$$EKE_w(\theta, \phi) = \cos(\theta) \left(\frac{u^{*2}(\theta, \phi) + v^{*2}(\theta, \phi)}{2} \right). \quad (1)$$

Here, we restrict our attention to approximately 440 m above ground level (agl), which serves as a representative height for low-level eddy activity.

Next, we average EKE_w zonally (across all longitudes) and meridionally between latitudes 20° – 77° of the hemisphere (at a fixed model level that approximately corresponds to 440 m agl height). Finally, we average over a given season (e.g. pre-solstice) to calculate the mean extratropical EKE denoted by $mEKE_v^{(j)}(\mathbf{exp}, k)$, where $\mathbf{exp}$ is the experiment-id (e.g. **control**), k is the year number of the experiment, $j = 1, 2$ corresponds to SH and NH, respectively, and $v = \alpha, \beta, \gamma$ corresponds to that hemisphere’s pre-solstice, solstice and post-solstice season, respectively. EKE_w should be interpreted as a point-wise area-weighted EKE, and $mEKE$ should be interpreted as the mean extratropical EKE in a given season. For each experiment and season, we calculate the mean, minimum, and maximum of $mEKE$ across the last 5 years of the simulation.

To explain the role of topography on hemispheric baroclinic wave activity asymmetry in MGCM simulations, we assess how this asymmetry is present across pre-solstice, solstice, and post-solstice seasons in the MGCM climate with the various topographic configurations. Similar to past studies (e.g., [Mulholland et al., 2016](#); [Lee et al., 2018](#)), we define the pre-solstice, solstice, and post-solstice seasons as the time intervals $L_s = 0$ – 60° , $L_s = 60$ – 120° , $L_s = 120$ – 180° (respectively) in the SH and

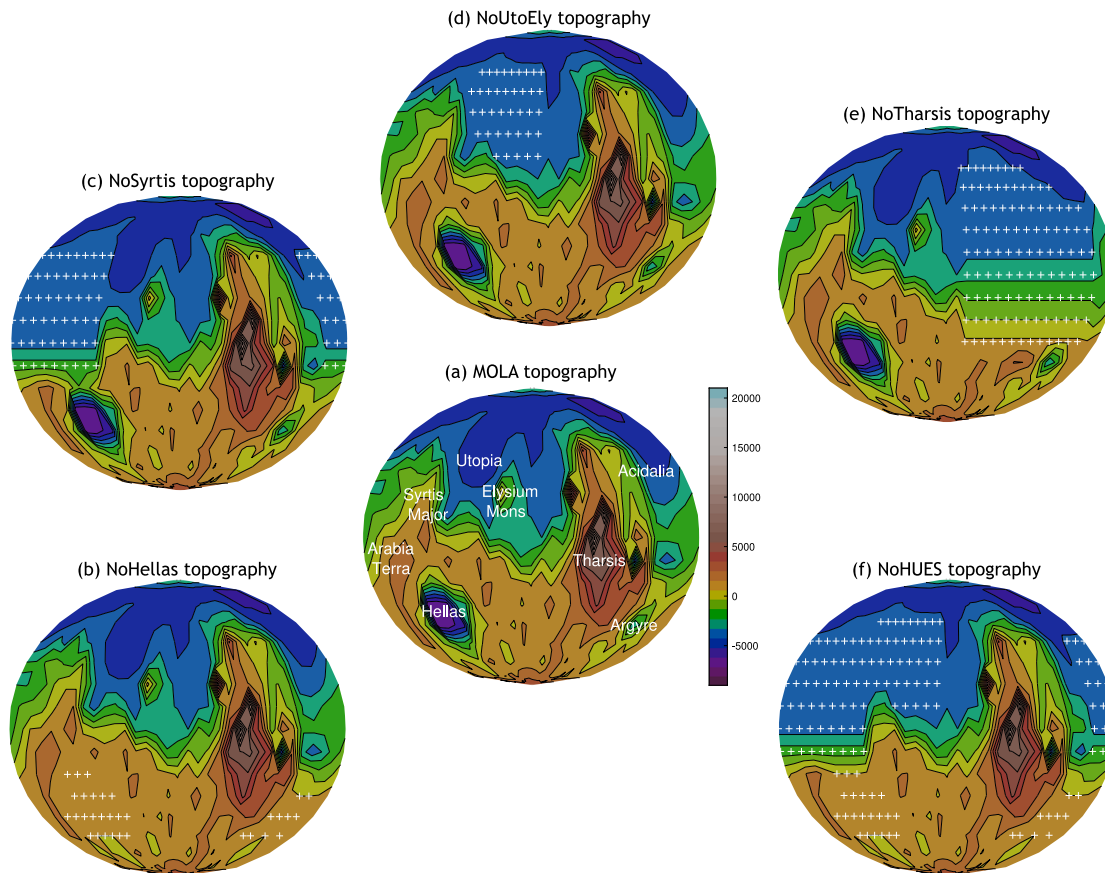


Fig. 1. MOLA topography (a), with key orographic regions named, and topographic configurations (in meters) that force experiments in group 2 (see Table 1) under Mollweide projection. Each experiment, white ‘+’ marker denotes regions where topographic manipulations were performed.

$L_s = 180\text{--}240^\circ$, $L_s = 240\text{--}300^\circ$, $L_s = 300\text{--}360^\circ$ (respectively) in the NH. While Lee et al. (2018) referred to these seasons as periods A, B, and C, we will refer to the pre-solstice, solstice, and post-solstice seasons as α , β , and γ seasons, respectively, to avoid confusion with the widely used A, B, and C dust storm period designations proposed by Kass et al. (2016).

We also point out that model results show intense wave activity in both hemispheres continuing into the post-equinox season ($L_s = 180\text{--}240^\circ$ in the SH and $L_s = 0\text{--}60^\circ$ in the NH). This wave activity is even stronger than the post-solstice season wave activity in several experiments. This post-equinox wave activity has not yet been observed directly in the SH, and recent developments suggest that such activity is not present in the SH (Hinson, 2025), although post-equinox activity is observed in the NH (see e.g., Hinson and Wilson, 2021; Sánchez-Lavega et al., 2025). However, the post-equinox wave activity is present in various Mars general climate models in both hemispheres, including Mars Weather Research and Forecasting (MarsWRF) GCM (Fig. 5 in Lee et al., 2018), Geophysical Fluid Dynamics Laboratory (GFDL) MGCM (Fig. 4 in Wilson et al., 2006), UK-MGCM (Fig. 3 in Mulholland et al., 2016) and in the LMD MGCM (Fig. 19 in Pottier et al., 2017). Post-equinox wave activity is also present in the reanalysis product Ensemble Mars Atmosphere Reanalysis System (EMARS) (Fig. 8d in Greybush et al., 2019) in years when data were supplied by MCS onboard the Mars Reconnaissance Orbiter (MRO). However, SH post-equinox wave activity is not present in the years when data were provided by the TES onboard Mars Global Surveyor (MGS). Accordingly, SH post-equinox wave activity is not captured by the reanalysis dataset Mars Analysis Correction Data Assimilation (MACDA) (Mulholland et al., 2016) that only covers years with TES data. Given the robust occurrence of SH post-equinox wave activity in various models, the absence of observations makes it unclear whether models exhibit a shared modeling bias or whether

detection methods fail to observe post-equinox season waves in the SH. Discussing the post-equinox activity for the NH without considering the SH would not be in the targeted scope of this study. Hence, in this work we refrain from discussing post-equinox wave activity, as this topic is planned for future work. In this paper, we focus on the pre-solstice, solstice, and post-solstice seasons.

3.1. Shortcomings of experiments with zonally averaged topography in assessing baroclinic wave activity

Our assessment of baroclinic wave activity started by first performing experiment **zonalmean** to compare and contrast with the results of Mulholland et al. (2016). However, we found that a **zonalmean** experiment is not as conclusive as perceived in the literature, and it might even be inconclusive. Namely, similar to the ‘Trough’ topography experiment of Matheson (2000), we found that a zonally averaged topography is not necessarily (perhaps depending on the details of the MGCM) applicable to assess the role of topographic zonal asymmetries on baroclinic waves, as our **zonalmean** experiment resulted in a strong transient channel flow during the solstice season in the latitude range of Hellas ($\sim 30\text{--}55^\circ\text{S}$, Fig. 2(a)). This resulted in the absence of a solstitial pause in the experiment (Fig. 3(a)) (as also observed in ‘wet’ conditions by Mulholland et al., 2016), however the lack of a pause is not a result of removing asymmetries, but the result of the artificial channel created by the zonal averaging of the Hellas basin. To seek a better representation of the effects of removing all zonal asymmetries, we compared experiments **zonalmean** and **zonalNoHellas**, where the topography is obtained by zonally averaging the **NoHellas** topography. Unsurprisingly, the channel flow at the latitude range of Hellas weakened significantly, albeit did not vanish entirely (Fig. 2(b)), and the

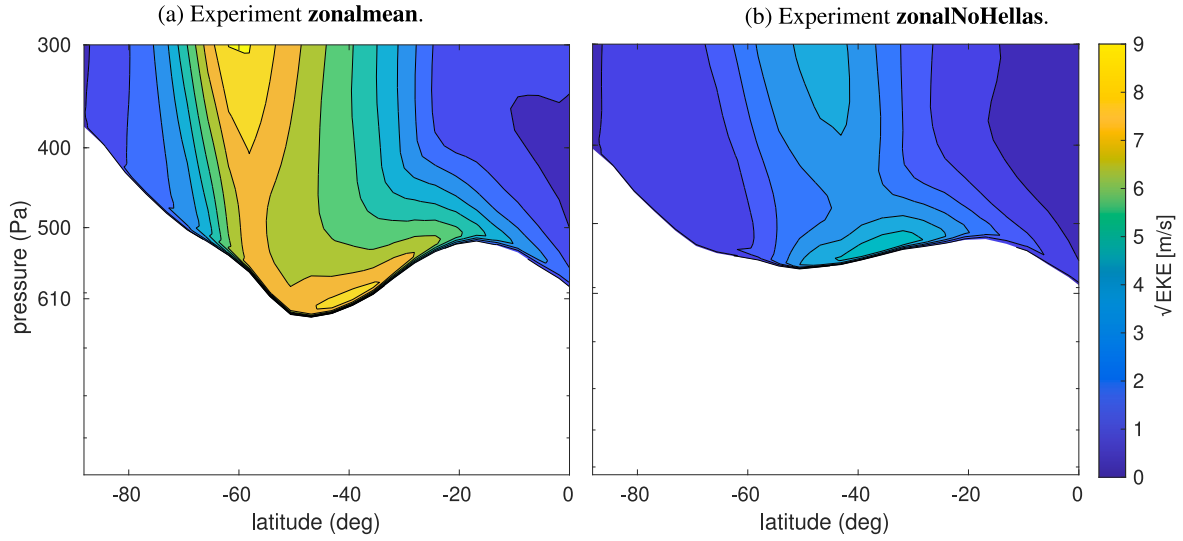


Fig. 2. Root of eddy kinetic energy (EKE) that is averaged zonally and seasonally during the SH winter solstice seasons ($L_s = 60\text{--}120^\circ$) of the 4th MY of experiments (a) **zonalmean**, (b) **zonalNoHellas**.

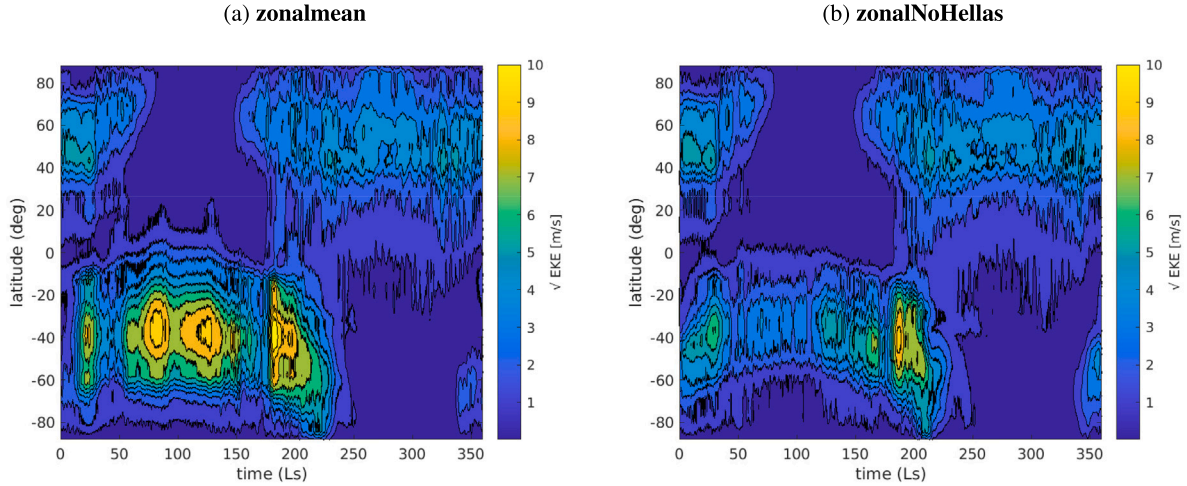


Fig. 3. Seasonal and spatial variations of the root of the zonally averaged area-weighted EKE (EKE_w) in experiments (a) **zonalmean** and (b) **zonalNoHellas**. While **zonalmean** lacks a solstitial pause, the pause reappears in **zonalNoHellas** when the channel flow is removed.

solstitial pause reappeared (Fig. 3(b)). Hence, the lack of a solstitial pause in **zonalmean** was not due to the removal of zonal asymmetries but because of the artificial generation of a strong channel flow in our experiments.

Although we have observed these topographic subtleties in our experiments, it might not be the case that in every Mars GCM experiment, including the simulations of Mulholland et al. (2016), a zonally averaged topography would lead to a strong channel flow. However, the lack of a solstitial pause resulted from a channel flow in our experiments; therefore, we conclude that the method of determining topographic effects through comparisons with a zonally averaged topography is not a suitable approach, as the resulting eddy activity is either strongly MGCM dependent or is amplified by a channel flow, undermining the investigation. Hence, we investigate baroclinic wave activity by considering various topographic configurations, where orographic regions are subject to targeted removal.

3.2. Topographic impacts on baroclinic wave activity

The influence of topography on $mEKE$ for the experiments illustrated in Fig. 4(a) (control, **NoHellas**, **NoTharsis**, **NoSyrtis**,

NoUtoEly, and **NoHUES**) is discussed next. Extratropical root mean EKE expressed little interannual variability (see error bars on Fig. 4(a)), suggesting that the 5-year average of a simulation using annually-repeating dust forcing is representative of any individual year (in terms of seasonally averaged EKE). A solstitial minimum in EKE, in each hemisphere, is present across all experiments. Therefore, our results suggest that topographic zonal variations may not be responsible for the solstitial pause. Each experiment with modified NH topography resulted in a weakening in NH $mEKE$ in all three (α, β, γ) seasons (Fig. 4(a), all panels except **NoHellas**). Hence, the NH wave activity is weakened if either a highland region is removed (**NoTharsis**, **NoSyrtis**, **NoUtoEly**) or a lowland region is landfilled (**NoUtoEly**), i.e., targeted, local removal of zonal asymmetries of NH topography has resulted in a consistent decrease in wave activity. By contrast, landfilling the impact basins in the SH (**NoHellas**) has substantially increased SH wave activity, i.e., local removal of zonal asymmetries in topography has increased wave activity. While the topographic alterations were aimed at removing specific components of orography leading to local reduction of zonal asymmetries in the targeted hemisphere's topography, the topographic modifications in experiments **NoTharsis**,

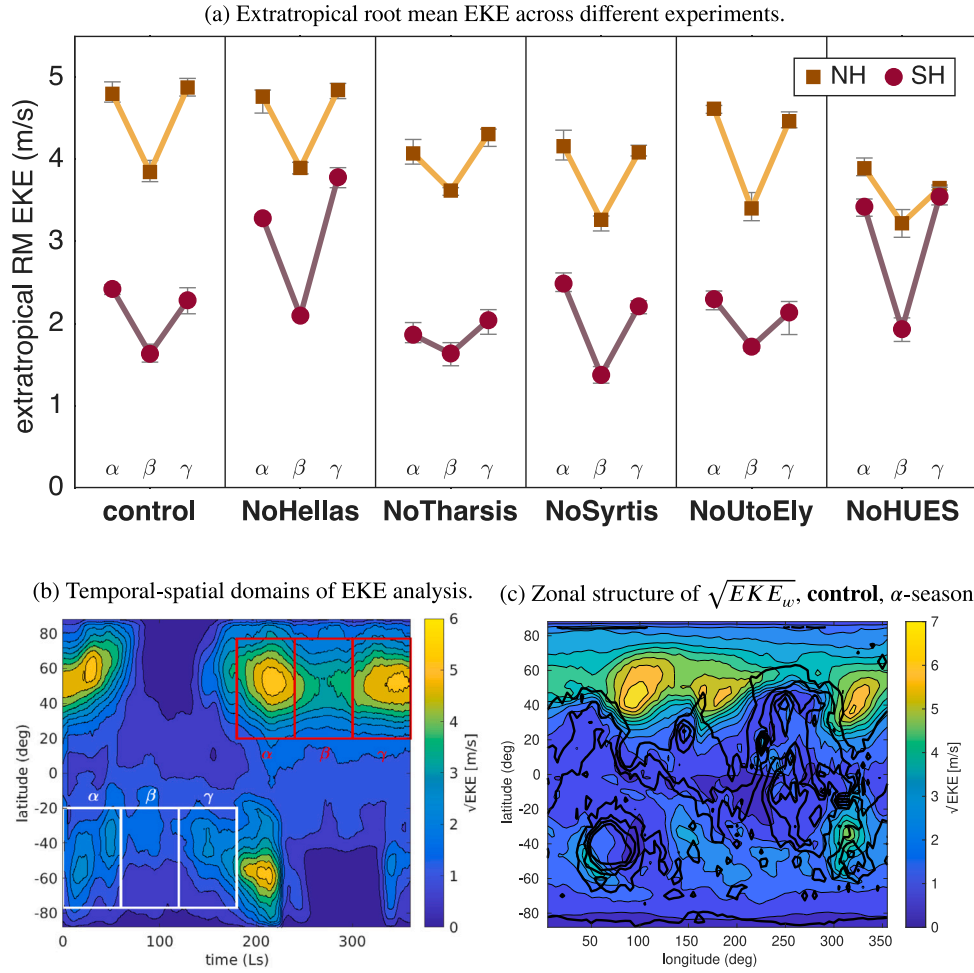


Fig. 4. (a) Comparison of extratropical root mean eddy kinetic energy ($\sqrt{mEKE}$) across experiments **control**, **NoHellas**, **NoTharsis**, **NoSyrtis**, **NoUtoEly**, **NoHUES** in the pre-solstice (α), solstice (β) and post-solstice (γ) seasons. For each experiment and season, the minimum, maximum, and mean values of $mEKE$ are calculated for the last 5 years of the experiment; filled squares (NH) and circles (SH) show the square root of mean $mEKE$ with error bars depicting minimum and maximum $\sqrt{mEKE}$. (b) Construction of $mEKE$ is illustrated using the EKE_w climatology from **control**. Red (NH) and white (SH) rectangles show the temporal-spatial domains where EKE_w is averaged across pre-solstice (α), solstice (β), and post-solstice (γ) seasons to calculate $mEKE$. (c) Root-mean EKE_w in experiment **control** is depicted. Area-weighted EKE (EKE_w) at each hemisphere is averaged through the hemisphere's pre-solstice season (that is, NH is averaged through $L_s = 180-240^\circ$, SH is averaged through $L_s = 0-60^\circ$) of the last 5 years of experiments **control**, at 440 m agl height. Bold lines represent topographic contours in **control**.

NoSyrtis also extended through the tropics into the SH, resulting in changes in wave activity in the SH.

These ‘cross-hemispheric’ influences are opposite for **NoTharsis**, where pre- and post-solstice season EKE (in the SH) is suppressed and solstice season EKE is unchanged, and for **NoSyrtis**, where solstice season EKE is weakened. In contrast, pre-solstice and post-solstice EKE did not change significantly. Therefore, the experiments show no consistent cross-hemispheric impact across different seasons.

While the experiments indicate a strong topographic influence on the zonally and seasonally averaged EKE (i.e., $mEKE$), it is impossible to determine from variations in $mEKE$ across different seasons and experiments (Fig. 4(a)) how topographic alterations modify the spatial structure of wave activity. Hence, in what follows, we establish locations of high wave activity with realistic topography and then determine how the spatial structure of EKE reacts to various topographic modifications.

3.2.1. High wave activity regions in simulations with realistic topography

We determine locations of high wave activity under realistic topography by averaging the area-weighted eddy kinetic energy, EKE_w , through the entire pre-solstice season in the last 5 years of experiment **control**. Accordingly, the averaging occurs at different time intervals

on the two hemispheres: NH is averaged through $L_s = 180-240^\circ$, and the SH is averaged through $L_s = 0-60^\circ$. When a different season (solstice or post-solstice) is considered, the EKE distributions may change quantitatively; however, we have not determined qualitative differences. That is, high EKE regions coincide across different seasons. For this reason, we restrict our attention to pre-solstice seasons here.

As also shown previously in several studies (e.g., Hollingsworth and Barnes, 1996; Wilson et al., 2006; Lewis et al., 2016; Mooring and Wilson, 2015; Battalio et al., 2016), Northern Hemisphere EKE amplification occurs (Fig. 4(b)) at the northwest parts of the Utopia, Arcadia, and Acidalia lowland regions that are located on the lee side of elevated regions Syrtis Major, Elysium Mons, and Tharsis, respectively. This amplification can be explained by lee cyclogenesis (Greybush et al., 2013), when low-level cyclonic motions can be triggered by the eastward traveling air column that expands vertically on the east side (downstream) of the highland regions, generating relative vorticity to conserve its potential vorticity. Downstream (eastward) of the lowlands, EKE is disrupted by the subsequently elevated regions: the Utopia-focused EKE is weakened over Elysium Mons, the Arcadia-related EKE is blocked by Tharsis, and Arabia Terra disrupts the Acidalia-focused EKE. Such a distribution of EKE suggests that elevated regions have both a blocking and a generating effect; EKE is

(a) Stationary meridional wind at $L_s = 60^\circ$, 46S in experiment **control**. (b) Stationary meridional wind at $L_s = 60^\circ$, 46S in experiment **NoHellas**.

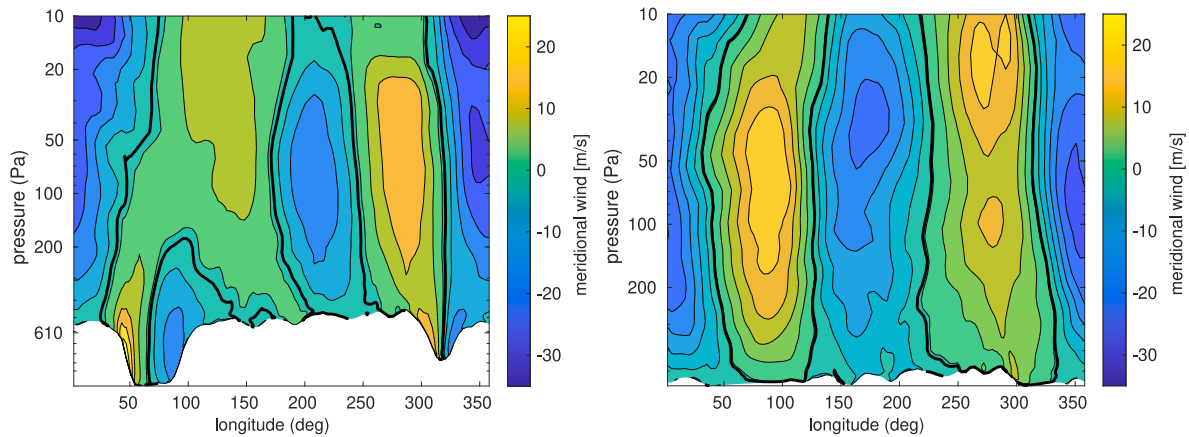


Fig. 5. Stationary meridional winds at $L_s = 60^\circ$, 46S, in **control** (a) and **NoHellas** (b). Bold lines represent zero levels in meridional wind.

weakened over high topography, but it strengthens in the subsequent (lee-side) downslope region. This pattern has been pointed out in reanalysis data by Battalio et al. (2016), who argued that the northern planitias (Utopia, Arcadia, Acidalia) are regions of high baroclinicity, and they are preceded upstream by a stationary wave forcing above the highland areas, and the downstream development of EKE is driven by geopotential flux convergence and eddy kinetic energy advection. Consequently, the downstream development is not a source of EKE but an energy transfer between different locations. Hence, it is anticipated that removing the highlands would weaken EKE amplification above the lowlands, which takes energy away from downstream development. However, such removal would be expected to lead to the disappearance of the stationary wave forcing above the highlands, providing an energy source for baroclinic wave growth above the highland structure (Yu and Hartmann, 1995). In the SH, EKE is focused on the western rims of the continental-scale basins Hellas (southwest rims) and Argyre (northwest rims) impact basins, where baroclinic waves are blocked by the stationary cyclonic circulations occupying the impact basins (Fig. 5(a)). The energetics of the blocking can be interpreted as was done by Battalio et al. (2018) using reanalysis data. Accordingly, the blocking is accompanied by baroclinic wave growth at the western vortex edge, initiated by strong vertical ageostrophic motions. However, baroclinic waves are suppressed within the vortex as baroclinic energy conversion is reduced. Furthermore, the mild baroclinic wave activity at the southern edge of the vortex can be explained through eddy kinetic energy transport.

Notably, topographic highland-lowland east–west structures modify NH wave activity, with Fig. 4(a) suggesting a net amplification. In the SH, impact basins suppress wave activity by blocking, but the question remains if they have any non-local influence on EKE. Next, we examine various topographic configurations to gain a deeper understanding of how they influence total baroclinic wave activity.

3.2.2. Impacts of Hellas and Argyre on baroclinic wave activity in the Southern Hemisphere

The existence of stable, long-lived, stationary cyclonic circulations within the continental-scale SH impact basins, Hellas and Argyre, is revealed by the **control** experiment which has realistic topography (Fig. 5(a)). These cyclonic circulations, or ‘gyres,’ form through the conservation of absolute vorticity (Kahre et al., 2020) as the column of air entering from the west elongates vertically, increasing relative vorticity, hence decreasing planetary vorticity through absolute vorticity conservation. These gyres develop in the pre-solstice season, and while Argyre’s gyre is constrained close to the surface above Argyre,

Hellas’ gyre extends to several km height above the crater rims (Fig. 5(a)) of Hellas; it can even extend to the lower reaches of the SH polar vortex at 100–150 Pa.

The removal of impact basins in **NoHellas** results in the disappearance of the atmospheric gyres over Hellas and Argyre (Fig. 5(b)). Instead, the stationary wind field is dominated by a deep, barotropic, wavenumber-2 stationary circulation, replacing the gyre-dominated wavenumber-3 near-surface atmospheric circulation. The northward branches of the resulting barotropic wave-2 stationary circulation are located above the land-filled Hellas and on the western flank of the land-filled Argyre, in agreement with the modeling work of Ogohara and Satomura (2008), who also found that the northward branch of the stationary circulation expands upon the entire region of the land-filled Hellas. Correspondingly, the lower atmosphere of an impact-basin-free Southern Hemisphere would fundamentally be characterized with a stationary wavenumber-2 wave structure. The formation of the impact basins results in localized stationary wave forcings above the craters that disrupt the wavenumber-2 structure and modify it into one dominated by a wavenumber-3 pattern. This simulation is associated with an increase in baroclinic wave activity everywhere in the 40–60°S latitude band, apart from a weakening at the western rims of the former craters (Fig. 6(b)) where EKE was amplified in **control**.

The most significant EKE increase is located in the eastern parts of the land-filled Hellas and Argyre, where EKE had been significantly reduced by blocking in **control**. The regions in the SH with the highest EKE, with a similar magnitude, are located at ~50E, ~150E, and ~300E. However, these regions are still weaker in EKE than Arcadia, Acidalia and Utopia (Fig. 6(a)). Furthermore, the SH high EKE region is more latitudinally constrained than the EKE regions at the NH lowlands. Apart from these high EKE locations (Acidalia, Utopia, Arcadia, and the Argyre-Hellas track), EKE shows similar intensity in the two hemispheres within the mid-latitude storm tracks. Consequently, while removing the impact basins has resulted in a hemispheric increase in baroclinic wave activity, the highest intensity regions in the SH cannot match the intensity of the lowlands in the NH, and the hemispheric asymmetry in wave activity persists, although weakened. Hence, the impact basins only account for a fraction of the hemispheric asymmetry.

3.2.3. Impacts of highlands and lowlands on baroclinic wave activity in the Northern Hemisphere

The removal of highland regions has resulted in an increase in EKE above the former highland region and a decrease in the lowland region downstream in each experiment, with the reduction seemingly carried

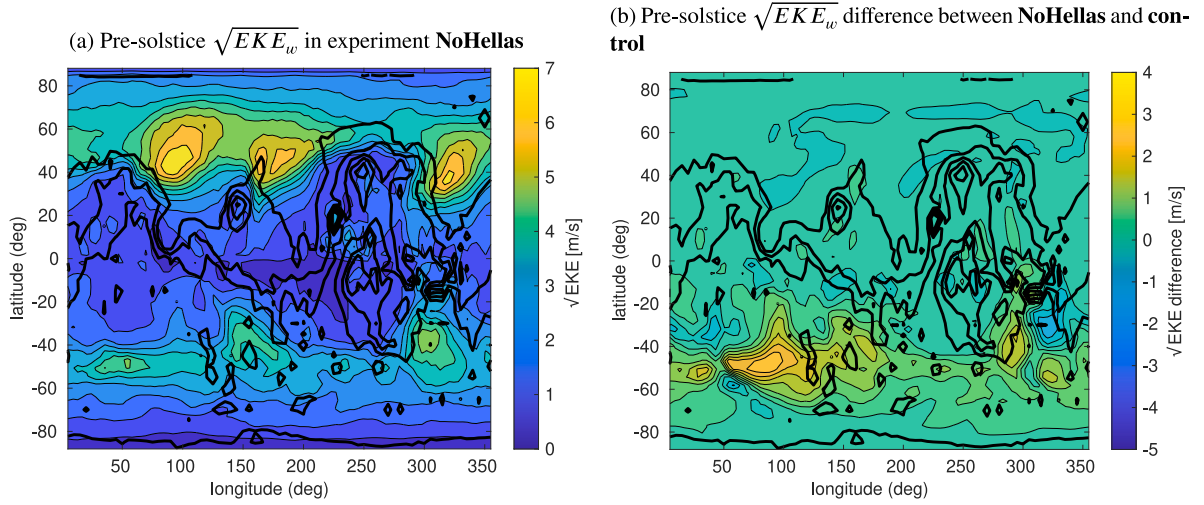


Fig. 6. Area-weighted EKE (EKE_w) at each hemisphere is averaged through the hemisphere's pre-solstice season (that is, NH is averaged through $L_s = 180-240^\circ$, SH is averaged through $L_s = 0-60^\circ$) of the last 5 years of experiments **NoHellas**, **control**. Root-mean EKE_w in experiment **NoHellas** is shown in (a), and the difference between root mean EKE_w of **NoHellas** and **control** is shown in (b). Thick black lines represent topographic contours corresponding to **NoHellas**.

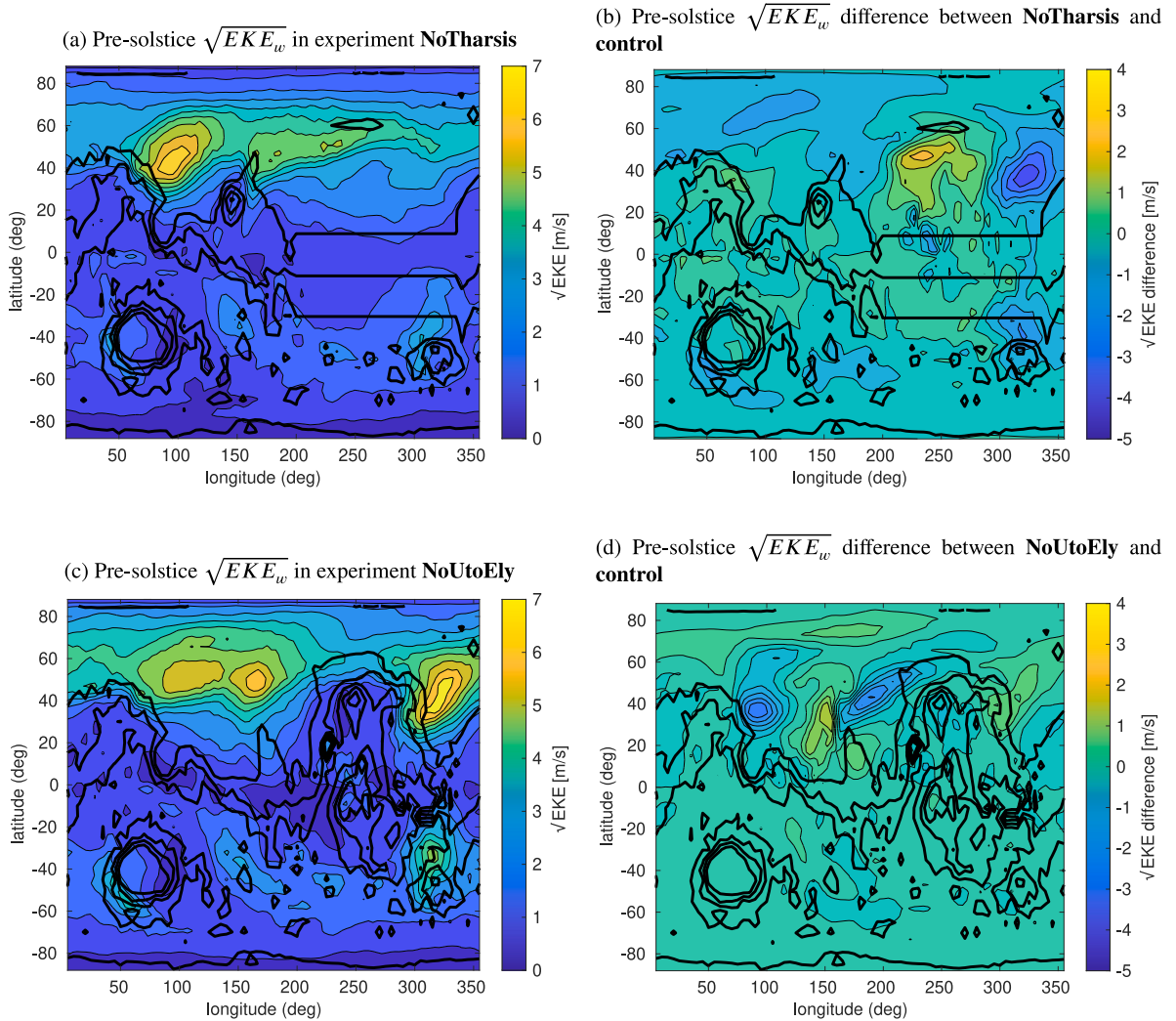


Fig. 7. Area-weighted EKE (EKE_w) at each hemisphere is averaged through the hemisphere's pre-solstice season (that is, NH is averaged through $L_s = 180-240^\circ$, SH is averaged through $L_s = 0-60^\circ$) of the last 5 years of experiments **NoTharsis**, **NoUtoEly**, and **control**. Root-mean EKE_w in experiment **NoTharsis** is shown on (a), the difference between root mean EKE_w of **NoTharsis** and **control** is shown on (b), root-mean EKE_w in experiment **NoUtoEly** is shown on (c), the difference between root mean EKE_w of **NoUtoEly** and **control** is shown on (d). Thick black lines represent topographic contours corresponding to **NoTharsis** on (a, b) and to **NoUtoEly** on (c, d).

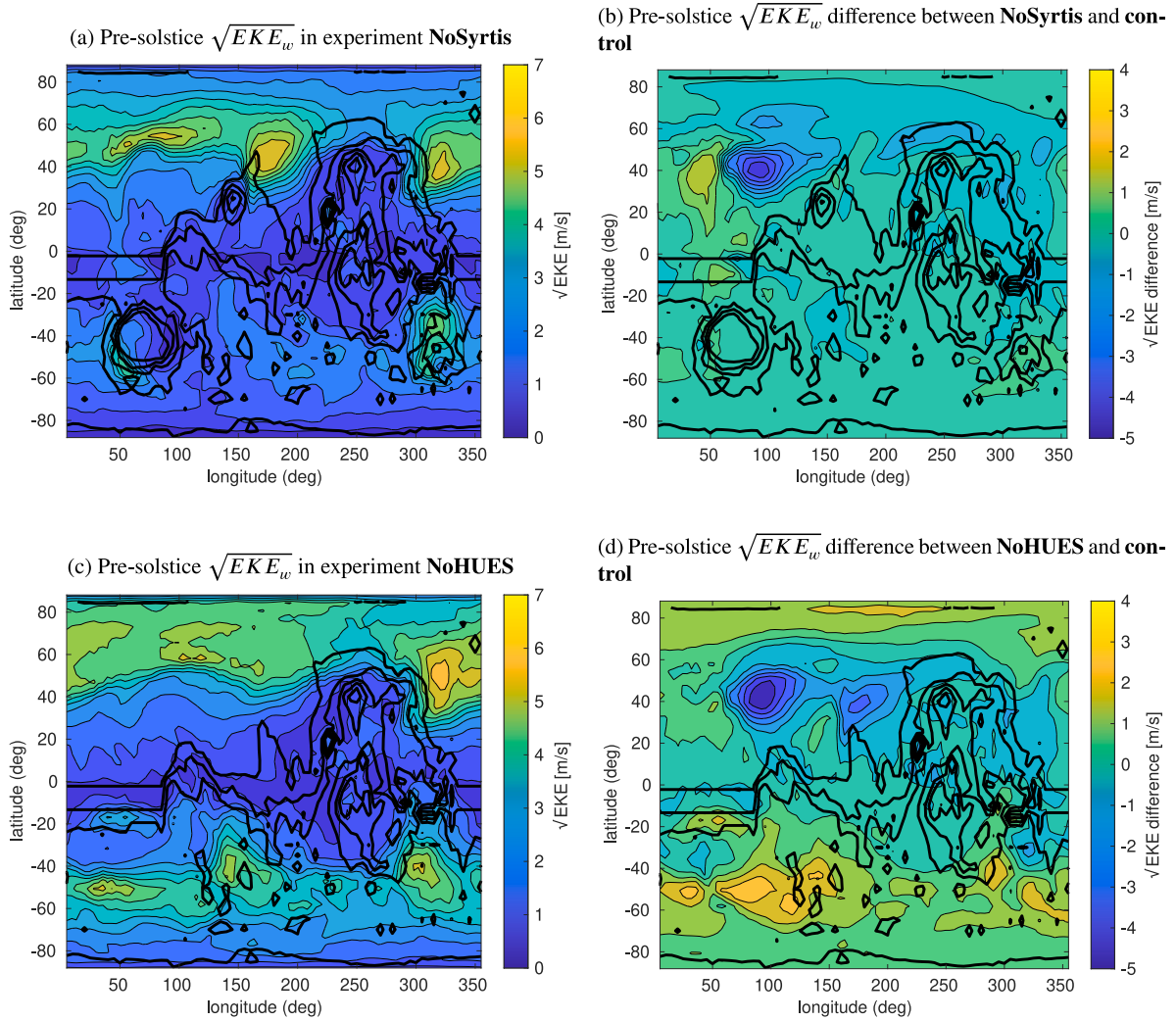


Fig. 8. Area-weighted EKE (EKE_w) at each hemisphere is averaged through the hemisphere's pre-solstice season (that is, NH is averaged through $L_s = 180 - 240^\circ$, SH is averaged through $L_s = 0 - 60^\circ$) of the last 5 years of experiments **NoSyrtis**, **NoHUES** and **control**. Root-mean EKE_w in experiment **NoSyrtis** is shown on (a), the difference between root mean EKE_w of **NoSyrtis** and **control** is shown on (b), root-mean EKE_w in experiment **NoHUES** is shown on (c), the difference between root mean EKE_w of **NoHUES** and **control** is shown on (d). Thick black lines represent topographic contours corresponding to **NoSyrtis** on (a, b) and to **NoHUES** on (c, d).

downstream until the next continental-scale highland region. Namely, the removal of Tharsis in experiment **NoTharsis** resulted in an increase in EKE above Tharsis, a weakening above Acidalia which lasts until Syrtis Major (Figs. 7(a), 7(b)), the removal of Utopia and Elysium in experiment **NoUtoEly** has resulted in a decrease in EKE above Utopia (Figs. 7(c), 7(d)), an increase above Elysium, and in a decrease above Arcadia (which is downstream of Elysium) that lasts until Tharsis. Finally, the removal of Arabia Terra and Syrtis Major in experiment **NoSyrtis** has increased EKE above Arabia Terra and in a decrease above Utopia, the decrease weakening downstream and extending to Elysium (Figs. 8(a), 8(b)).

Hence, in Mars' realistic NH climate, highland/lowland east-west structures result in a weakening in EKE above highlands (with a stationary circulation taking up wave energy in place), a strengthening above lowlands through lee cyclogenesis, with the strengthening being carried downstream until the next highland, resulting in a net gain in EKE. By removing each of Arabia Terra, Utopia, Elysium, Hellas, and Argyre in **NoHUES** (Figs. 8(c), 8(d)), both NH amplification and SH wave suppression are decreased sufficiently so that the hemispheric asymmetry in wave activity disappears in the post-solstice season and comes to a near-balance in the pre-solstice season (Fig. 4(a)). However,

solstice-season balance is not achieved, and topography alone does not seem to explain hemispheric wave asymmetry.

4. Discussion

This study has made two main contributions to the research area of baroclinic wave activity on Mars. The first contribution is the identification of potential channel wave formation at the latitude band of Hellas in simulations with zonally averaged topography. Thus, SH baroclinic wave amplification in experiment **zonalmean** is not a result of the lack of zonal variations, but rather, a consequence of channel formation. Therefore, it is not a suitable method to assess the effect of topography on baroclinic wave activity by forcing the MGCM with a zonally averaged topography. This issue has a particularly significant impact on solstitial wave activity; the channel wave in our experiment is strongest during the southern solstice season and is responsible for the lack of solstitial pause. This responsibility is revealed when the MGCM is run with a topography that is zonally averaged after Hellas and Argyre are land-filled; in this case the solstitial pause reappears. Furthermore, each of our experiments has resulted in a solstitial pause,

in each hemisphere, suggesting that the solstitial pause is not a result of topographic forcing.

Second, the key outcome of this study is that topographic suppression influences the hemispheric baroclinic wave asymmetry in the SH and amplification in the NH. The SH suppression is attributed to the development of stable atmospheric gyres in the mid-latitude impact basins, and the NH amplification is attributed to the massive highland-lowland structures at NH mid-latitudes. While it has been shown in reanalysis output (Lewis et al., 2016; Battalio et al., 2018) that wave activity is blocked at Hellas, it has not been addressed how the stationary wave circulation would change and how intense extratropical traveling waves would be in the absence of impact basins. Our simulations show that land-filling the impact basins does not necessarily lead to hemispheric wave symmetry. Numerous works have pointed out that the NH wave activity is focused within the lowlands (Hollingsworth and Barnes, 1996; Wilson et al., 2006; Mooring and Wilson, 2015; Lewis et al., 2016; Battalio et al., 2016). Yet, no one has analyzed whether these topography-induced baroclinic wave structures manifest as a net EKE gain or loss. Overall, we find that by removing NH highland-lowland structures and SH impact basins, near-symmetry of the hemispheric baroclinic wave activity can be achieved in both pre- and post-solstice seasons, although balance has not been achieved in the solstice seasons.

A few alternative sources that can account for the hemispheric asymmetry, either qualitatively or quantitatively, may be noted. First, we considered eddy kinetic energy per unit mass in this study rather than per unit volume. This decision was motivated by its prevalence in literature focusing on energetics in Mars' baroclinic waves (see e.g., Battalio et al., 2018). However, if eddy kinetic energy per unit volume were considered instead, quantitative differences in the results would be expected, as the density at a fixed above ground level height would vary both spatially (with higher densities corresponding to topographic lows) and seasonally (with the atmospheric density reacting to changes in the CO₂-cycle). Second, since our experiments did not include radiatively active water ice clouds and were forced by a tame dust scenario, it is possible that aerosol forcing can account for further hemispheric wave asymmetry. Determination of that is encouraged in future work. Furthermore, we did not assess the influence of the polar ice caps on baroclinic wave activity. It is noteworthy to point out that the surface frost distribution within the NH ice cap changes across experiments, since anticyclones suppress and cyclones enhance ice cap expansion. This modification of the ice cap boundary could lead to secondary effects that enhance/suppress baroclinic wave growth in certain regions. In particular, EKE growth observed polewards of 77°N in NoUtoEly (Fig. 7(d)) and NoHUES (Fig. 8(d)) might relate to such differences in the NH ice cap.

5. Conclusions

Several computational experiments were conducted using the NASA Ames MGCM with various topographic configurations to investigate the role of Mars' topography in its hemispheric asymmetry in baroclinic wave activity. We demonstrated that while the impact basins in the SH are occupied by strong, stable vortices that significantly suppress local baroclinic wave activity, this suppression is insufficient to explain the hemispheric asymmetry in wave activity. Additionally, we found that net wave activity in the NH is amplified by its topography. Consequently, the hemispheric asymmetry in transient wave activity is driven by both SH wave suppression and NH wave enhancement. In particular, we established that NH highland-lowland west-east structures result in a suppression of EKE over the highland region (accompanied by the appearance of a stationary structure) and an enhancement in EKE over the lowland region with the enhancement persisting further downstream, resulting in a net gain in seasonally averaged extratropical EKE in each of the pre-solstice, solstice and post-solstice seasons. Furthermore, we

have illustrated that the removal of the SH impact basins and localized removal of NH highland-lowland structures (experiment NoHUES) leads to the disappearance of the hemispheric asymmetry in wave activity in the pre- and post-solstice seasons. However, topographic influences alone could not explain hemispheric wave asymmetry between the two solstice seasons.

CRediT authorship contribution statement

Márton Mester: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **R. John Wilson:** Writing – review & editing, Software, Investigation, Conceptualization. **Jeffery L. Hollingsworth:** Writing – review & editing, Conceptualization. **Melinda A. Kahre:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Here, we provide descriptions of the topographic setups used in our experiments. We denote MOLA topography by $\tau(\lambda, \theta)$ on a longitude-latitude grid. The topographic configurations $h(\lambda, \theta)$ used in the different experiments are provided below, brackets $\langle \cdot \rangle$ denote zonal average, topographic height units are in [m]:

- **control:** realistic topography, i.e. $h(\lambda, \theta) = \tau(\lambda, \theta)$.
- **zonalmean:** realistic topography zonally averaged, i.e. $h(\lambda, \theta) = \langle \tau(\lambda, \theta) \rangle$.
- **NoHellas:** Hellas and Argyre are filled up to 1 km in height.

$$h(\lambda, \theta) = h_{nohellas}(\lambda, \theta) := \begin{cases} 1000 & \text{if } \theta \in (-66, -17)N, \lambda \in (33, 114)E, \tau(\lambda, \theta) < 1000, \\ 1000 & \text{if } \theta \in (-64, -34)N, \lambda \in (252, 340)E, \tau(\lambda, \theta) < 1000, \\ \tau(\lambda, \theta) & \text{otherwise.} \end{cases} \quad (2)$$

- **zonalNoHellas:** zonally averaged 'NoHellas' topography, $h(\lambda, \theta) = \langle h_{nohellas}(\lambda, \theta) \rangle$.
- **NoTharsis:** Tharsis is removed and replaced by a meridional dual-slope structure,

$$h(\lambda, \theta) = \begin{cases} 1800 - 104 \cdot (\theta + 40) & \text{if } \theta \in (-40, 10)N, \lambda \in (200, 340)E, \\ -3300 - 15 \cdot (\theta - 10) & \text{if } \theta \in (10, 60)N, \lambda \in (200, 340)E, 5\lambda + 9\theta < 1988, \\ \tau(\lambda, \theta) & \text{otherwise.} \end{cases} \quad (3)$$

- **NoUtoEly:** Utopia is filled up and Elysium Mons is removed,

$$h(\lambda, \theta) = \begin{cases} -3600 & \text{if } \theta \in (5, 60)N, \lambda \in (78, 170)E, 6\lambda + 11\theta > 835, \\ \tau(\lambda, \theta) & \text{otherwise.} \end{cases} \quad (4)$$

- **NoSyrtis:** Syrtis and Arabia Terra are removed,

$$h(\lambda, \theta) = \begin{cases} -3600 & \text{if } \theta \in (0, 50)N, \lambda \in (335, 90)E \bmod 360, \\ -3600 + 180 \cdot \theta & \text{if } \theta \in (-20, 0)N, \lambda \in (335, 90)E \bmod 360, \\ \tau(\lambda, \theta) & \text{otherwise.} \end{cases} \quad (5)$$

- **NoHUES:** Hellas, Argyre, Utopia filled up, Syrtis Major, Arabia Terra, Elysium Mons removed,

$$h(\lambda, \theta) = \begin{cases} 1000 & \text{if } \theta \in (-66, -17)N, \lambda \in (33, 114)E, \tau(\lambda, \theta) < 1000, \\ 1000 & \text{if } \theta \in (-64, -34)N, \lambda \in (252, 340)E, \tau(\lambda, \theta) < 1000, \\ -3600 & \text{if } \theta \in (5, 60)N, \lambda \in (78, 170)E, 6\lambda + 11\theta > 835, \\ -3600 & \text{if } \theta \in (0, 50)N, \lambda \in (335, 90)E \bmod 360, \\ -3600 + 180 \cdot \theta & \text{if } \theta \in (-20, 0)N, \lambda \in (335, 90)E \bmod 360, \\ \tau(\lambda, \theta) & \text{otherwise.} \end{cases} \quad (6)$$

- **NoAcidalia:** Southern Acidalia and Chryse are removed,

$$h(\lambda, \theta) = \begin{cases} -675 & \text{if } \theta \in (-20, 40)N, \lambda \in (290, 50)E \bmod 360, \\ \tau(\lambda, \theta) & \text{otherwise.} \end{cases} \quad (7)$$

Data availability

The NASA Ames MGCM version 3.1 is publicly available at doi.org/10.25966/ty4s-w176. The low-level zonal and meridional wind output from the **control**, **NoHellas**, **NoTharsis**, **NoSyrtis**, **NoUtoEly**, and **NoHUES** NASA Ames MGCM experiments, and electronic versions of **Figs. 2** and **5**, are archived and publicly available at 10.5281/zenodo.16658505.

Dataset link: <https://zenodo.org/records/16658505>.

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